

Digital simulation workshops to enhance student learning of Chemical Engineering Thermodynamics

Mauro Luberti, Thomas L. Rodgers & Percy van der Gryp

To cite this article: Mauro Luberti, Thomas L. Rodgers & Percy van der Gryp (2026) Digital simulation workshops to enhance student learning of Chemical Engineering Thermodynamics, Interactive Learning Environments, 34:3, 1748-1762, DOI: [10.1080/10494820.2025.2529445](https://doi.org/10.1080/10494820.2025.2529445)

To link to this article: <https://doi.org/10.1080/10494820.2025.2529445>



© 2025 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group



[View supplementary material](#)



Published online: 27 Jul 2025.



[Submit your article to this journal](#)



Article views: 1190



[View related articles](#)



[View Crossmark data](#)



Citing articles: 1 [View citing articles](#)



OPEN ACCESS



Digital simulation workshops to enhance student learning of Chemical Engineering Thermodynamics

Mauro Luberti^a, Thomas L. Rodgers^a and Percy van der Gryp^{a,b}

^aDepartment of Chemical Engineering, School of Engineering, The University of Manchester, Manchester, UK;

^bDepartment of Chemical Engineering, Faculty of Engineering, Built Environment and Information Technology, University of Pretoria, Pretoria, South Africa

ABSTRACT

Chemical Thermodynamics is a notoriously challenging course in Chemical Engineering programmes where several students struggle to grasp the rather abstract concepts and relate them to practical applications. At the same time, this subject has been highly impacted by digitalization as more complex thermodynamic models and simulation techniques have become increasingly available. This study evaluated the benefits of integrating digital simulation workshops grounded in Experiential Learning Theory (ELT) within our first-year Chemical Engineering Thermodynamics course at the University of Manchester. The workshops were designed to use the commercial Aspen Properties software as a rapid vapour–liquid equilibrium (VLE) generator. The workshops covered numerous aspects of VLE, including the selection of appropriate thermodynamic models such as equations of state and activity coefficient models, and the assessment of various diagrams for ideal, real and azeotropic mixtures, as well as high-pressure mixtures. A survey was carried out to gather the students' opinions about the workshops structure, technical content, and perceived usefulness. It was found that our digital workshops coupling the use of interactive simulations with a problem-based learning approach not only promoted deeper engagement and critical thinking but also improved students' academic performance in the final exam and their overall attitude towards the course.

ARTICLE HISTORY

Received 5 March 2025

Accepted 29 June 2025

KEYWORDS

Digitalization;
Thermodynamic models;
Vapour–liquid equilibrium;
Aspen Properties;
Experiential learning theory

1. Introduction

Chemical Engineering Thermodynamics is at the heart of Chemical Engineering as it dictates the ultimate boundaries for chemical reactors and separation processes. One of its main goals is to provide quantitative estimates of equilibrium properties of mixtures as required for chemical process design. The importance of teaching Chemical Thermodynamics in higher education arose soon after the birth of Chemical Engineering as a discipline in 1887 in Manchester. In fact, in 1920s it was already taught as a separate course in some universities in the USA while in 1930s it became a clear pre-requisite for Unit Operations courses (Wankat, 2009). More recently, Chemical Thermodynamics has been identified as one of the “Six Pillars of Chemical Engineering” (McCarthy & Parker, 2004) and several higher education institutions have proposed different courses ranging from

CONTACT Mauro Luberti  mauro.luberti@manchester.ac.uk

 Supplemental data for this article can be accessed online at <https://doi.org/10.1080/10494820.2025.2529445>.

© 2025 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group

This is an Open Access article distributed under the terms of the Creative Commons Attribution-NonCommercial-NoDerivatives License (<http://creativecommons.org/licenses/by-nc-nd/4.0/>), which permits non-commercial re-use, distribution, and reproduction in any medium, provided the original work is properly cited, and is not altered, transformed, or built upon in any way. The terms on which this article has been published allow the posting of the Accepted Manuscript in a repository by the author(s) or with their consent.

BEng to PhD level (Kontogeorgis et al., 2009). Currently, there are also well-established textbooks on this topic (Elliott & Lira, 2012; Prausnitz et al., 1999; Sandler, 2017; Smith et al., 2022), covering chemical equilibrium, vapour–liquid equilibrium, gas–liquid equilibrium, liquid–liquid equilibrium, solid–liquid equilibrium, and less conventional phase equilibria such as adsorption (Santori et al., 2015; Santori & Luberti, 2016). Beyond its importance, Chemical Thermodynamics has long received the reputation of being a notoriously challenging course for students because of its rather abstract concepts, problem-solving nature and associated mathematical modelling. Nevertheless, these features have prompted the shift from traditional, teacher-centred classrooms towards more active, student-centred environments through experiential learning and digital technologies (Le et al., 2022).

According to the Experiential Learning Theory (ELT), students create knowledge from practical experience rather than just from received instruction (Bergsteiner et al., 2010), which, in the context of higher education, translates into problem-solving sessions, project-based activities, in-class questions, etc. For instance, Calvo and Prieto (2016) discussed a teaching methodology involving computer-based labs to solve advanced distillation processes within a fourth-year course in Chemical Engineering. Dong (2021) proposed a set of options in a second-year Thermodynamics course for encouraging student participation and engagement. The activities consisted of in-class questions and research-oriented group projects having the additional goals of promoting oral and written communication skills.

At the same time, digital technologies are impacting how we work and what we work on, and this change affects our economies and societies. The role of digitalization in Chemical Engineering has dramatically increased since computers became broadly available in the 1960s. In the last two decades multi-criteria optimization has been successfully applied to the development of thermodynamic models including equations of state, activity coefficient models and force-field molecular models (Barba et al., 2015; Forte et al., 2019). In higher education, digital technologies have allowed a dynamic and flexible learning experience through, for instance, an active interaction with mathematical models via adjusting parameters and watching the responses. In this context, Bansagi and Rodgers (2018) developed a user-friendly and accessible web-based graphical tool for teaching ternary diagrams and liquid–liquid extraction to Chemical Engineering students. Furthermore, Rodgers et al. (2023) presented a set of online interactive graphs to support the student learning of unit operations in a more engaging way. Bale and Singh (2023) successfully demonstrated how a novel virtual method based on Simtronics SPM 700 can be utilized in Chemical Engineering Unit Operation Laboratory to teach distillation processes to undergraduate students. Caserta et al. (2021) applied an interactive smartphone-based student response system in large classroom settings of an undergraduate Chemical Engineering Thermodynamics course showing high effectiveness in student learning and allowing early detection of struggling students. Bascunana et al. (2023) developed and implemented in a Chemical Engineering Master's programme a set of interactive Jupyter Notebook activities related to mass balances, reactor design, and separation operations. As a result, student engagement and participation improved along with student marks in the final exam.

Recent works in the literature have highlighted the synergy between experiential learning and digital technologies in teaching Chemical Engineering Thermodynamics. In particular, complex multi-component thermodynamic models could be relatively easily coded using a programming software (Excel, MATLAB, Octave, Python, Mathcad, Mathematica, Polymath) or readily accessed with a process simulation software (Aspen Plus, Aspen HYSYS, UniSim, PRO/II, CHEMCAD, gPROMS, ProSim, DWSIM). In addition, there are a number of thermophysical property libraries often available free and open source equipped with state-of-the-art as well as high-accuracy property formulations. These include, for instance, CoolProp based on C++ programming language (Bell et al., 2014) and PYroMat fully embedded in Python interface (Martin et al., 2022).

Castier and Amer (2011) showed the capabilities of a free, open-code Excel add-in software called XSEOS for use in Chemical Engineering Thermodynamics courses involving phase and chemical equilibrium calculations. The tool allowed the calculation of thermodynamic properties, such as residual

and excess properties as well as fugacity and activity coefficients, for pure components and mixtures. Additional capabilities were the characterization of petroleum fractions and the computation of surface tensions. The authors claimed that student feedback on the software usage improved over the years mainly because of the availability of a more detailed manual and the higher experience of instructors with the package. Martin et al. (2011) presented a set of tools for incorporating practical exercises using complex equations of state in an Applied Thermodynamics course. The thermodynamic models included cubic equations of state and group-contribution equations of state, such as SAFT and PC-SAFT. The programmes were coded in both MATLAB and Octave environments to be easily reusable by means of an object-oriented methodology. A case study based on a course at Master's level demonstrated that the use of these programming tools has had a generally positive impact on student learning, especially in the understanding of the theoretical aspects of the course. Suthar and Joshipura (2023) developed an integrative approach of simulation-based exercises to increase student engagement in a second-year Chemical Engineering Thermodynamics course. The exercises included the selection of appropriate thermodynamic models, the assessment of vapour–liquid equilibrium (VLE) via diagrams and the estimation of binary interaction parameters for the correct prediction of VLE. The exercises were implemented using two open-source and CAPE-OPEN-compliant process simulators, COCO and DWSIM. In addition, a student survey revealed that the simulation exercises have been overall very helpful to understand the theory concepts and the wider applications to other core Chemical Engineering courses. Paoli et al. (2024) designed a short course on basic statistical thermodynamic theories and advanced thermodynamic modelling tools for upper undergraduate and postgraduate students. The adopted software package was an open-source thermodynamic toolkit called Clapeyron, which was implemented in Julia language. The course consisted of three sections covering the thermodynamic fundamentals, the implementation of thermodynamic models and their use for solving practical phase equilibrium problems. After a pilot study the students found the material very engaging while the educators positively highlighted the modular nature of the course content that enabled them to easily customize the problems.

In this work, we presented digital simulation workshops aligned with ELT within our first-year Chemical Engineering Thermodynamics course at the University of Manchester. The workshops were designed to improve student learning combining interactive simulations with a problem-based approach. We then evaluated the benefits of integrating these workshops through a dedicated student survey based on their academic performance in the final exam. Although several authors emphasized the importance of developing free and open-source modelling software we implemented the workshops using the commercial Aspen software as a teaching tool because of the following invaluable advantages:

- The Aspen Suite (AspenTech, 2024) is the most widespread chemical process simulator worldwide in both academia and industry (Schefflan, 2016). Thus, it is most likely that our students will be using it in their future jobs as process engineers.
- There exists a substantial amount of supporting resources on the web, via textbooks and research papers (Borreguero et al., 2019; Calvo & Prieto, 2016);
- Aspen software can be readily integrated into multiple courses within the Chemical Engineering programmes, including Chemical Process Simulation, Process Control, Process Optimization, Process Integration, Design Projects, Dissertations, etc. At the University of Manchester, it is in fact fully integrated into the Chemical Process Simulation and Design Project courses, so students will be using Aspen again in their second and third year of studies.

2. Context

The Chemical Engineering Thermodynamics course at the University of Manchester is taught in the second semester of the first year of the Chemical Engineering programmes. This course is undertaken by all first-year students which are typically 170–200 students. The course is delivered over

11 weeks via blended learning with asynchronous material delivered weekly to students via the University's virtual learning environment, Blackboard. This includes recorded lecture videos, interactive tasks, progress quizzes and discussion boards. The online asynchronous content is then supported by weekly 2-hour long synchronous in-class sessions where students are provided with theory recaps and complements, practical applications, step-by-step solutions of past week problems as well as active learning and problem-based work on the current week problems under the supervision of the unit leaders and the graduate teaching assistants (GTAs). The intended learning outcomes (ILOs) and weekly contents of the Chemical Engineering Thermodynamics course can be found in [Table 1](#).

The digital simulation workshops were introduced during the second semester of academic year 2023–2024. Two 2-hour workshops were run during weeks 7 and 9, in parallel to the delivery of the course, and designed specifically to support the topics related to vapour–liquid equilibrium (Contents 4–7). This is because VLE is the most important phase equilibrium in Chemical Engineering and typically occupies a large content fraction in a first Chemical Thermodynamics course. Moreover, the digital workshops were associated to ILOs 2, 3 and 5 ([Table 1](#)). In 2023–2024, 145 students out of 176 attended the digital workshops, corresponding to approximately 82% of the overall student cohort.

3. Digital simulation workshops

The Chemical Thermodynamics workshops were designed based on ELT principles with the main focus of maximizing student active learning and critical thinking. The workshops were organized such that students gathered in a large PC-cluster and worked individually, in pairs or in small groups, following instructions available on Blackboard under the supervision of the unit leaders and the GTAs. Furthermore, students had the chance to ask questions to a staff member anytime during the workshops while the unit leaders would occasionally clarify certain points to the whole student cohort. The overall structure and contents of the Chemical Engineering Thermodynamics workshops are summarized in [Table 2](#) and full details are reported in the *Supplementary Information*.

It should be noted that the examples used in the workshops in terms of binary systems and thermodynamic models were intentionally chosen to be relatively simple as the workshops were designed for first-year students learning Chemical Thermodynamics for the first time. It should also be noted here that although students became skilled in using Aspen Properties software, the main aim was developing their understanding of VLE. Based on the comments above, Aspen Properties was a very useful and appropriate rapid VLE generator.

Table 1. Intended learning outcomes (ILOs) and contents of Chemical Engineering Thermodynamics course.

ILO	Description
ILO 1	Use the knowledge of intermolecular forces to qualitatively predict the properties of bulk matter
ILO 2	Apply equations of state to predict the properties of one-component systems and binary mixtures
ILO 3	Apply activity coefficient models to predict the properties of liquid binary mixtures
ILO 4	Manipulate the equations governing chemical equilibria so as to model chemical engineering processes
ILO 5	Solve and critically evaluate phase and chemical equilibria analytically and by means of computer-based modelling programs
Content	Description
Content 1	Intermolecular forces and macroscopic vs microscopic
Content 2	"Free" energies and Maxwell relations
Content 3	Chemical potential and phase equilibrium
Content 4	Equations of state and residual properties
Content 5	Chemical potential of real fluids and excess properties
Content 6	Fundamentals of vapour–liquid equilibrium
Content 7	Vapour–liquid equilibrium calculations
Content 8	Fundamentals of gas–liquid equilibrium
Content 9	Fundamentals of liquid–liquid equilibrium
Content 10	Chemical reaction equilibrium

Table 2. Structure and contents of Chemical Engineering Thermodynamics workshops. In square brackets are reported workshop session and task duration. Full details can be found in the *Supplementary Information*.

Content	Binary system	Model	Main tasks	Additional tasks
• Recap of VLE formulation and selection of thermodynamic models [W1, 30 min]				
• Getting started with Aspen Properties [W1, 45 min]				
• VLE of pure component [W1, 15 min]	Pure methane	Peng-Robinson	1. Plot P - T diagram	i. Identify critical point coordinates ii. Discuss
• VLE at low pressure				
○ Ideal mixtures [W1, 30 min]	benzene(1)/toluene(2)	Ideal solution	1. Plot P - x - y and y - x diagrams at 25°C 2. Plot T - x - y diagram at 1.0 bar	i. Identify P_1^S and P_2^S ii. Identify T_1^B and T_2^B iii. Discuss
○ Real mixtures [W2, 30 min]	acetone(1)/water(2)	Wilson	1. Plot P - x - y and y - x diagrams at 35°C 2. Plot T - x - y diagram at 0.2 bar	i. Identify P_1^S and P_2^S ii. Identify T_1^B and T_2^B iii. Type of deviation? Discuss iv. Azeotrope? Composition? v. Re-plot P - x - y diagram at 85°C vi. Replot all diagrams using ideal solution model, discuss
○ Azeotropic mixtures [W2, 30 min]	acetone(1)/chloroform(2)	Non-random two-liquid	1. Plot P - x - y and y - x diagrams at 10°C 2. Plot T - x - y diagram at 2.0 bar	i. Identify P_1^S and P_2^S ii. Identify T_1^B and T_2^B iii. Type of deviation? Discuss iv. Azeotrope? Composition? v. Re-plot P - x - y diagram at 160°C vi. Replot all diagrams using Peng-Robinson model, discuss
• VLE at high pressure				
○ Normal behaviour [W2, 30 min]	propane(1)/n-butane(2)	Soave-Redlich-Kwong	1. Plot the P - T diagram at 50/50 mol%	i. Identify mixture critical point coordinates ii. Retrograde behaviour? Criconderbar and criconderthem coordinates? iii. Discuss
○ Retrograde behaviour [W2, 30 min]	carbon dioxide(1)/n-hexane(2)	Soave-Redlich-Kwong	1. Plot the P - T diagram at 66.7/33.3 mol%	i. Identify mixture critical point coordinates ii. Retrograde behaviour? Criconderbar and criconderthem coordinates? iii. Discuss

3.1. Recap of VLE formulation and selection of thermodynamic models

The first section of the workshops was intended as a recap of the VLE formulation and the selection of thermodynamic models. This was mainly to scaffold the workshop sessions and link them to the course asynchronous material, making sure that students understood they were applying the theory they had learned. Moreover, the recap was particularly useful for those students who were not able to keep up with the asynchronous material and the synchronous in-class sessions.

Students were reminded that the formulation of VLE is based on pressure and type of mixture.

At low pressures, VLE is formulated with the γ - ϕ approach:

$$\phi_i^V y_i P = \gamma_i x_i \phi_i^S P_i^S \phi_i^P \quad (1)$$

where P is the pressure, and, for the generic component i : y_i is the mole fraction in the vapour phase, ϕ_i^V is the fugacity coefficient in the vapour phase, P_i^S is the vapour pressure, x_i is the mole fraction in the liquid phase, γ_i is the activity coefficient, ϕ_i^S is the fugacity coefficient in liquid phase at saturation and ϕ_i^P is the Poynting correction factor. In Eq.(1) ϕ_i^V and ϕ_i^S can be calculated using any equation of state while γ_i needs be calculated using an activity coefficient model.

For sufficiently low pressures Eq.(1) can be simplified in the modified Raoult's law as follows:

$$y_i P = \gamma_i x_i P_i^S \quad (2)$$

Additionally, if the mixture is constituted of chemically similar components with close boiling point ($<30^{\circ}\text{C}$) Raoult's law holds:

$$y_i P = x_i P_i^S \quad (3)$$

At high pressures, VLE is formulated with the ϕ - ϕ approach:

$$\varphi_i^V y_i = \varphi_i^L x_i \quad (4)$$

where φ_i^L is the fugacity coefficient in the liquid phase. In Eq.(4) φ_i^V and φ_i^L need to be calculated using a cubic equation of state.

In addition, students were introduced to the concept of "decision tree" in order to select the appropriate thermodynamic models depending on pressure and type of mixture, and were shown the detailed decision tree from Aspen based on Carlson's influential article (Carlson, 1996). However, for the purpose of the workshops, students could also refer to a simplified decision tree, as depicted in Figure 1.

3.2. Getting started with Aspen Properties

After the VLE recap, students were introduced to various simulation software used in the chemical industry with a focus on the Aspen Suite (Aspen Plus, Aspen HYSYS) and the subprogram Aspen Properties. Then, following instructions available on Blackboard (see *Supplementary Information*), they were guided step-by-step on how to access the software, create/save/open/close a case file, add components from the database and select the thermodynamic models. It was also explained in detail how to plot VLE diagrams, including P - T diagrams for a pure component, P - x - y , T - x - y and y - x diagrams for binary systems at either fixed temperature or fixed pressure, and P - T diagrams for binary systems at fixed composition. In addition, it was shown how to modify the component order, change the units of measurement, format the graphics of the diagrams, export the diagrams, etc.

3.3. VLE of pure component

As a warm-up exercise, students were tasked to plot the P - T diagram for pure methane using the Peng-Robinson model. It was requested to identify and note down the critical point coordinates (P_C , T_C) and compare them with information available online and/or in the textbooks. As a follow-

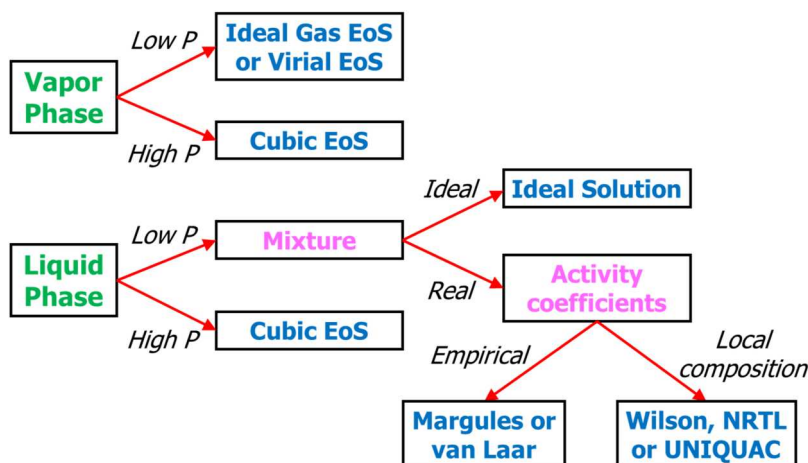


Figure 1. Simplified decision tree for thermodynamic model selection.

up, students were instructed to repeat the exercise for higher hydrocarbons (e.g. ethane, propane, n-butane, etc.) and explain how and why the critical point coordinates shifted.

3.4. VLE at low pressure

In this section of the workshops students studied the VLE of various binary systems at low pressure, including ideal mixtures, real mixtures and azeotropic mixtures.

3.4.1. Ideal mixtures

Using the ideal solution model students were tasked to plot the isothermal P - x - y and y - x diagrams at 25°C and the isobaric T - x - y diagram at 1.0 bar for the binary system benzene(1)/toluene(2). Given that the mixture is constituted of chemically similar components, students had to confirm that the resulting binary diagrams exhibited in Figure 2 were indeed associated to an ideal mixture. This could be done, for instance, by inspecting the bubble point curve in Figure 2a (P - x diagram) and noticing that it is linear. Additional tasks included the recording of the pure component vapour pressures and boiling temperatures from the P - x - y and T - x - y diagrams, respectively.

3.4.2. Real mixtures

Using the Wilson model students were tasked to plot the isothermal P - x - y and y - x diagrams at 35°C and the isobaric T - x - y diagram at 0.2 bar for the binary system acetone(1)/water(2). From chemistry knowledge both components are polar and exhibit hydrogen bonding, therefore it is expected that the mixture does not behave ideally. First, students had to assess the type of deviation of the mixture, including the presence of potential azeotropes. Then, they were instructed to re-plot the P - x - y diagram at 85°C to spot differences in the VLE behaviour. Key findings can be drawn from Figure 3 where it is clear that at 35°C the binary system exhibits positive deviations from Raoult's law due to the concavity of the bubble point curve, but no azeotrope is present (Figure 3a). By contrast, a maximum-pressure azeotrope at around $x_1 = y_1 = 0.86$ is formed when the temperature is raised to 85°C as a result of higher repulsive forces in the liquid phase (Figure 3b). As an additional task, students had to re-plot all previous diagrams using the ideal solution model and realize that it was not able to capture the deviations from ideality nor the presence of azeotropes. This is because the ideal solution model cannot predict any type of intermolecular interactions.

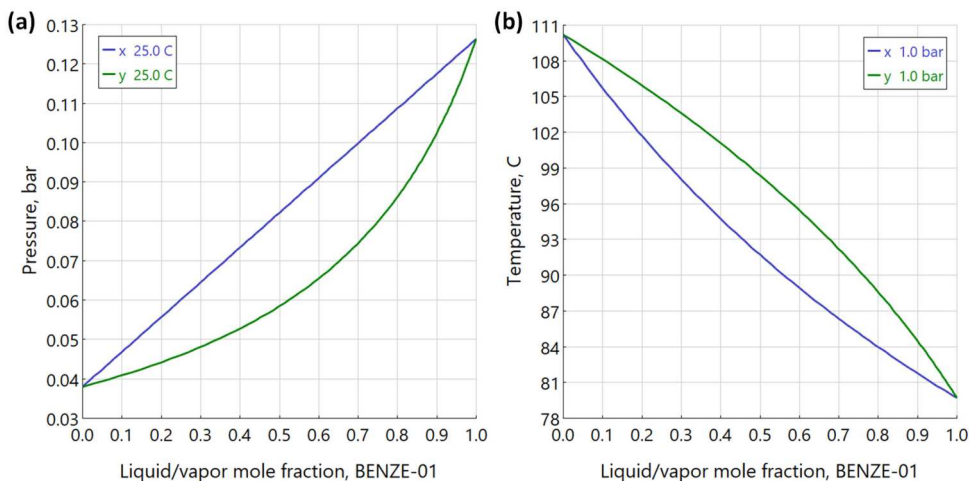


Figure 2. P - x - y diagram at 25°C (a) and T - x - y diagram at 1.0 bar (b) for the binary system benzene(1)/toluene(2) using the ideal solution model.

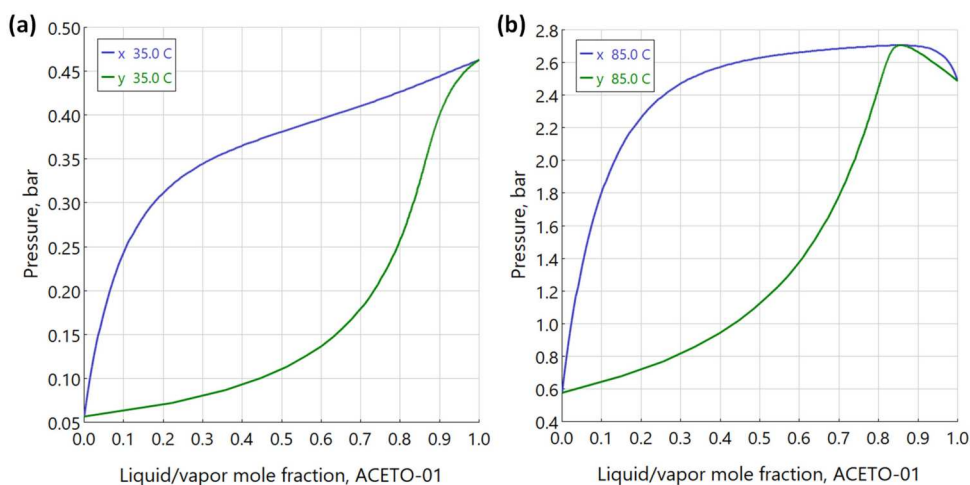


Figure 3. P - x - y diagram at 35°C (a) and 85°C (b) for the binary system acetone(1)/water(2) using the Wilson model.

3.4.3. Azeotropic mixtures

Using the non-random two-liquid model students were tasked to plot the isothermal P - x - y and y - x diagrams at 10°C and the isobaric T - x - y diagram at 2.0 bar for the binary system acetone(1)/chloroform(2). Also in this case both components are polar and are expected to form a real mixture. In particular, students had to realize that at 10°C the P - x - y diagram exhibits a minimum-pressure azeotrope at around $x_1 = y_1 = 0.38$ while it disappears at 160°C due to the lower attractive forces at higher temperatures. It was underscored that the removal of azeotropes by pressure or temperature variation will have important implications in the future separation courses. Eventually, students had to re-plot all previous diagrams using the Peng-Robinson model and realize that not only it was not able to capture the maximum-temperature azeotrope at 2.0 bar in the T - x - y diagram but that the boiling temperatures of pure components were not correctly estimated, as shown in Figure 4. This is because the Peng-Robinson model fails to predict the negative deviations and the azeotropes as the components in the mixture are polar.

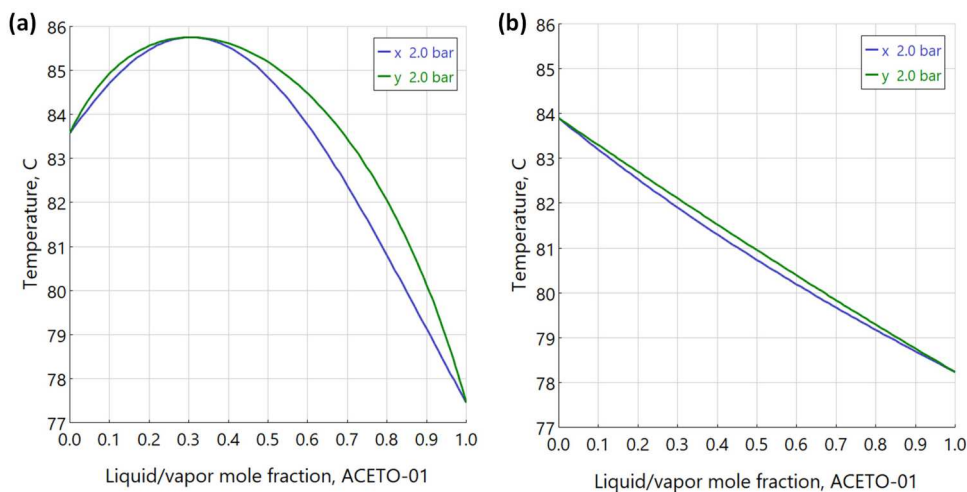


Figure 4. T - x - y diagram at 2.0 bar for the binary system acetone(1)/chloroform(2) using the non-random two-liquid model (a) and the Peng-Robinson model (b).

3.5. VLE at high pressure

In this section of the workshops students studied the VLE of various binary systems at high pressure, including those exhibiting normal behaviour and retrograde behaviour.

3.5.1. Normal behaviour

Using the Soave-Redlich-Kwong model students were tasked to plot the P - T diagram for the binary system propane(1)/n-butane(2) with a fixed composition of 50/50 mol%. The resulting diagram is shown in Figure 5a where the region between the two curves is bi-phasic. Students had to realize that the bubble point and dew point curves meet at the critical point that is also at the maximum pressure (≈ 42 bar) and maximum temperature ($\approx 130^\circ\text{C}$), thus no retrograde behaviour is observed for this mixture.

3.5.2. Retrograde behaviour

Using the same Soave-Redlich-Kwong model students were tasked to plot the P - T diagram for the binary system carbon dioxide(1)/n-hexane(2) with a fixed composition of 66.7/33.3 mol%. Differently from the previous case, this mixture exhibits retrograde behaviour of both 1st and 2nd kind given that the resulting critical point is not at the maximum pressure nor at the maximum temperature, as exhibited in Figure 5b. In fact, the cricondenbar is around 112 bar, thus higher than the critical pressure (108 bar), and the cricondentherm is around 170°C , therefore higher than the critical temperature (150°C). It should be noted that in Figure 5b the bubble point and dew point curves do not completely meet due to the lack of accuracy associated with the Soave-Redlich-Kwong model close to the critical point. Students were reminded that this is a common issue for most cubic equations of state.

4. Student survey and results

To evaluate the benefits of integrating the digital simulation workshops within the Chemical Engineering Thermodynamics course we employed a survey to gather the students' opinions on the workshop structure, technical content and perceived usefulness. We also assessed the students' academic performance in the final exam compared to the previous year.

4.1. Survey design

This study employed a paper survey that was handed out to students in the following synchronous in-class session right after the workshops so that they still had a vivid memory of the experience. The

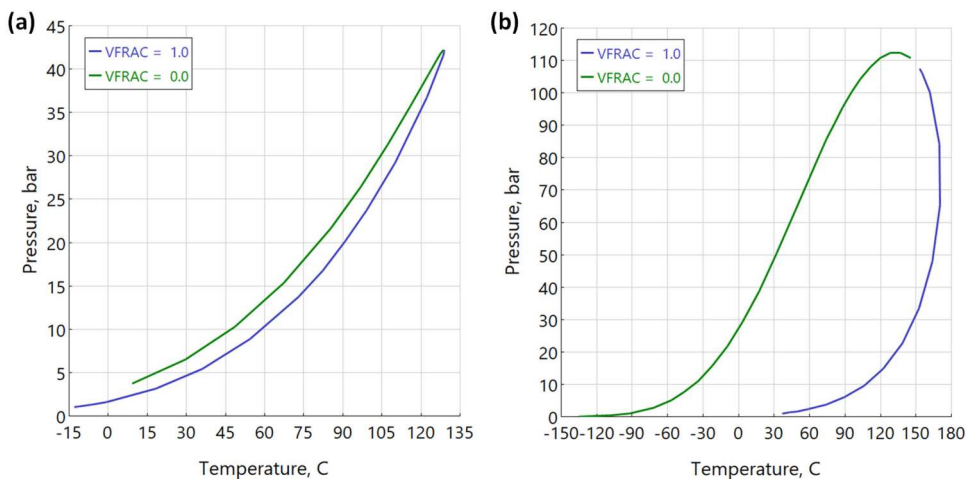


Figure 5. P - T diagram for the binary system propane(1)/n-butane(2) at 50/50 mol% (a) and carbon dioxide(1)/n-hexane(2) at 66.7/33.3 mol% (b) using the Soave-Redlich-Kwong model.

paper approach was selected from best practice, as this produces a higher number of student responses in comparison to an online survey. In addition, no incentive was given to the students, so the survey was filled out on a completely voluntary and anonymous basis.

The survey consisted of two sections. The first section focussed on student demographics such as their gender (male, female, other) and provenance (home, international). The second section comprised 11 questions designed to cover the workshop structure, technical content and perceived usefulness. Students had to indicate their agreement with each question on a five-point Likert scale, from 1 (completely disagree) to 5 (completely agree). At the bottom of the survey there was also an empty box where students could write suggestions for improving the workshops. The list of survey questions and statements can be found in [Table 3](#).

4.2. Student demographics

The answers of the survey were collected from 52 students corresponding to approximately 30% of the overall first-year student cohort (and about 36% of the students who physically attended the workshop). Out of the total, around 65% of students identified as males and 35% identified as females. In addition, around 56% were home students against 44% being international students. Student demographic information is reported in [Table 4](#). It should be noted that these figures were fully representative of the total class demographic.

4.3. Survey results

The results of the survey are summarized in [Table 5](#), detailing the Likert point count, average and standard deviation (STD) per each question, and exhibited in [Figures 6](#) and [7](#) based on the student percentage agreement with questions. In particular, [Figure 6](#) refers to questions concerning the workshop structure and perceived usefulness (Q1, Q2, Q10, Q11) while [Figure 7](#) refers to questions concerning the workshop technical content (Q3–Q9). Overall, the answers were positive for most aspects of the workshops, with Likert point averages in the range of 3.4–4.3 out of 5 ([Table 5](#)).

From [Figure 6](#) it is clear that the vast majority of students was very satisfied with the way the workshops were organized (Q1) and the perceived usefulness to consolidate their knowledge

Table 3. Survey on Chemical Engineering Thermodynamics workshops.

Question	Statement
1	Overall, the workshops were well organized
2	Overall, the workshops have consolidated my knowledge on vapour-liquid equilibrium
3	The workshops have improved my understanding of the use of equations of state
4	The workshops have improved my understanding of the use of activity coefficient models
5	I can now better characterize ideal mixtures at low pressures via P - x - y , T - x - y and y - x diagrams
6	I can now better characterize real mixtures at low pressures via P - x - y , T - x - y and y - x diagrams
7	I now better understand the concept of azeotrope
8	I can now better characterize mixtures at high pressures via P - T diagrams
9	I now better understand the concept of retrograde behaviour
10	I now feel more confident about the selection of thermodynamic models in the chemical industry
11	The understanding I have gained in the workshops will improve my performance on this course

Table 4. Student demographic information.

Characteristic	Item	Frequency	Percentage (%)
Gender	Male	34	65.4
	Female	18	34.6
	Other	0	0.0
Provenance	Home	29	55.8
	International	23	44.2
Total		52	100.0

Table 5. Overall results of the survey; 1: completely disagree, 2: disagree, 3: neutral, 4: agree, 5: completely agree.

Question	1	2	3	4	5	Average	STD
1	0	0	5	27	20	4.29	0.63
2	0	0	11	29	12	4.02	0.66
3	0	3	13	25	11	3.85	0.82
4	1	3	23	14	11	3.60	0.95
5	0	1	7	27	17	4.15	0.72
6	0	1	9	28	14	4.06	0.72
7	1	1	18	17	15	3.85	0.93
8	0	2	10	29	11	3.94	0.74
9	1	10	16	17	8	3.40	1.02
10	1	3	16	22	10	3.71	0.91
11	0	0	15	25	12	3.94	0.72

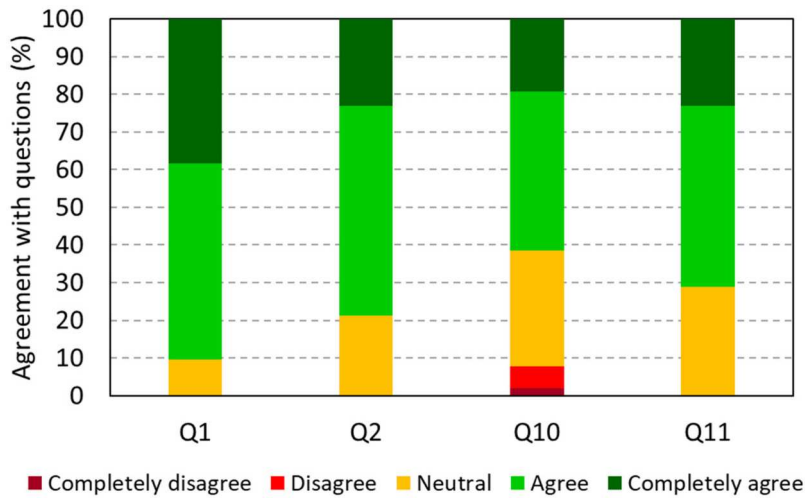


Figure 6. Survey results in relation to workshop structure and perceived usefulness based on student percentage agreement with questions.

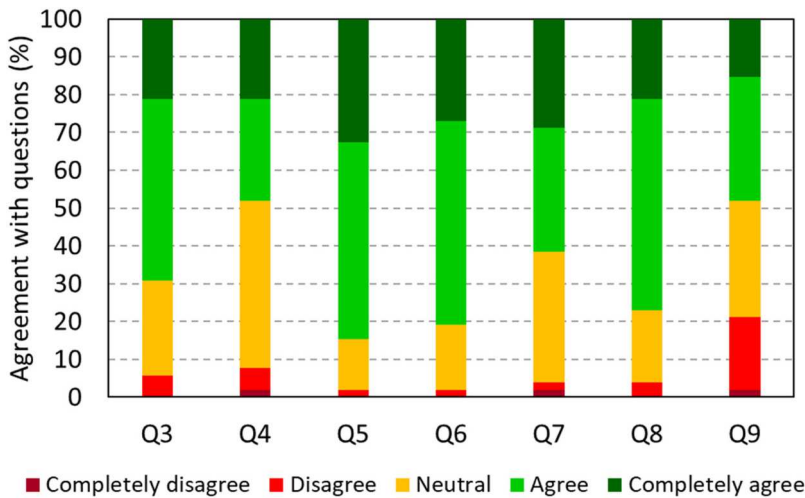


Figure 7. Survey results in relation to workshop technical content based on student percentage agreement with questions.

on VLE (Q2). Furthermore, students' feedback on their confidence in the selection of thermodynamic models (Q10) and their expected enhanced performance in the course (Q11) was rather favourable, as 62–71% of students agreed or strongly agreed (positive agreement) with these questions. This also demonstrated a marked improvement in the students' overall attitude towards the course.

From Figure 7 a student positive agreement of 78–85% was observed in relation to the ability of better characterizing ideal and real mixtures at low pressure via P - x - y , T - x - y and y - x diagrams and mixtures at high pressure via P - T diagrams, which was extensively covered in the workshops (Q5, Q6, Q8). As regards the understanding of equations of state (Q3) and activity coefficient models (Q4) the results were quite different: the first showed a student positive agreement of 69% while the latter showed a value of 48%. This could indicate a potential shortfall in the students' learning of the chemical potential of real mixtures in the liquid phase, which should be better explained and/or revisited. Nevertheless, a favourable result came from the understanding of the concept of azeotropes (Q7), revealing a student positive agreement of 62%. Azeotropes are notoriously challenging to grasp, especially in a first Thermodynamics course, but the interactive and graphical-based approach of the workshops has arguably helped the students assimilate this concept. The same conclusions could not be drawn for the concept of retrograde behaviour (Q9) where the student positive agreement was only 48%. However, it should be noted that VLE at high pressure was the last topic covered right before the workshops, so it is likely that students needed more reflection and retention time on this course content.

The survey results were also assessed based on students' demographics (gender and provenance) via the two-tail Student's t -test, which for Likert scales provides a good statistical representation regardless of distribution and number of samples (Norman, 2010; Sullivan & Artino, 2013). However, the resulting p -values were not statistically significant for any of the questions.

Figure 8 represents a summary of students' suggestions for improving the workshops. It can be seen that three students recommended to make a video to demonstrate some VLE diagrams in Aspen Properties before the workshop sessions. This suggestion is certainly beneficial, and we will implement it next year to improve the overall students' experience. Two students recommended to have more/clearer instructions as part of the online material. We will reflect upon providing more instructions as this could be possibly detrimental for the group – and problem-based approach. In addition, two other students requested to provide more details about the theory. We will aim to address this comment in the recap of VLE formulation and selection of thermodynamic models although this section was already designed to take 30 minutes, which is a sufficiently long time. There were also a couple of comments about solving the additional tasks and going

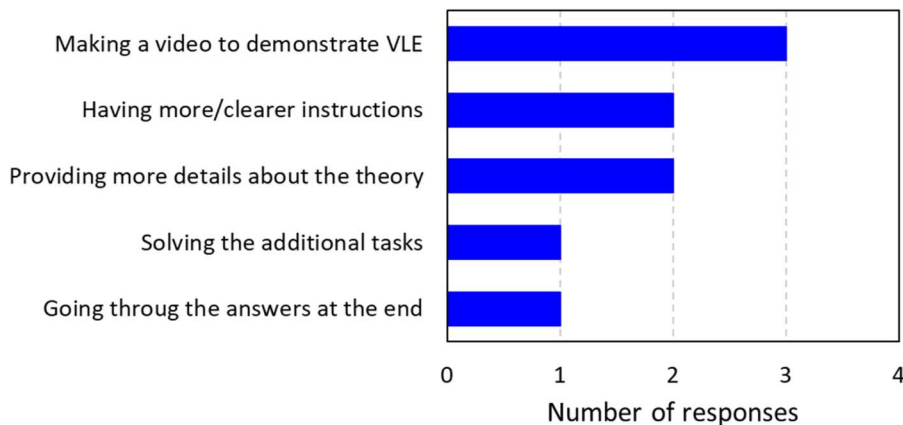


Figure 8. Summary of students' suggestions for improving the workshops.

Table 6. Statistics of student marks in final exam for academic years 2022–2023 (before workshops implementation) and 2023–2024 (after workshops implementation), including the two-tail Student's *t*-test with unequal variances and the right-tail *F*-test for the standard deviations. *is the result of the left-tail *F*-test.

Exam question	2022–2023		2023–2024		<i>p</i> (<i>t</i> -test) on Averages	<i>p</i> (<i>F</i> -test) on STDs
	Average	STD	Average	STD		
Exam Q1	59.2	25.0	65.6	22.2	0.012	0.055
Exam Q2	44.4	26.8	56.7	20.4	<0.001	<0.001
Exam Q3	72.0	21.6	74.2	22.1	0.35	0.38*

through the answers at the end of the workshops to consolidate the learning, however the majority of students was able to finish all the tasks well ahead of the scheduled time and left the PC-cluster. The detailed solutions of the workshops were posted online on Blackboard immediately after.

4.4. Student academic performance

The benefits of integrating the digital simulation workshops within the Chemical Engineering Thermodynamics course were further quantified by evaluating the statistics of student marks in the final exam in the academic year 2023–2024 compared to the previous academic year 2022–2023 when the workshops were not run. The final exam consisted of three questions, of which the first two were on phase equilibria, therefore closely linked to the workshops, and the third question on chemical reaction equilibria.

Table 6 reports the student average marks and standard deviations for each exam question as well as the *p*-values based on the two-tail Student's *t*-test with unequal variances (this was used as the worst-case scenario as the two-tail test will give higher *p*-values compared to the one-tail test) for the two academic years. The table also reports the *p*-values based on the right-tail *F*-test to see if the standard deviations have reduced (for Exam Q3 the standard deviation increased so a left-tail *F*-test was carried out). As can be seen from Table 6, the average marks increased for all three exam questions, especially those concerning phase equilibria (+6.4% in Exam Q1, +12.3% in Exam Q2). However, differently from Exam Q3, the results of both Exam Q1 and Exam Q2 were statistically significant as the *p*-values were much less than 0.05. In addition, for Exam Q1 and Exam Q2 the standard deviations reduced, almost statistically significantly to the 0.05 level and much less than 0.05, respectively. For Exam Q3 the standard deviation increased, but not by a significant amount. This showed that not only the average marks increased for the questions on phase equilibria, but students at the lower end increased more, thus reducing the attainment gap of the class. Overall, this confirmed that the workshops have had a positive effect on the students' learning of phase equilibria.

5. Conclusions

In this work, we presented and assessed digital simulation workshops integrated within our first-year Chemical Engineering Thermodynamics course at the University of Manchester. The workshops were designed on the principles of the Experiential Learning Theory and made use of digital technologies to maximize student active learning and critical thinking. The chosen teaching tool was Aspen software given its large use worldwide in both academia and industry, the vast availability of supporting resources and the capability to integrate it into other courses in the programme.

The workshops focussed on VLE and comprised numerous contents, including a recap of VLE formulation, the appropriate selection of thermodynamic models, and the plot of phase diagrams for pure component, ideal, real and azeotropic mixtures as well as high-pressure mixtures. Additional tasks enabled students to achieve a deeper understanding of the subject, such as the appearance/disappearance of azeotropes by pressure/temperature variation, the influence of thermodynamic model change on VLE, and the difference between normal and retrograde behaviour.

A dedicated survey was handed to students in order to gather their opinions on the workshop structure, technical content and perceived usefulness. Overall, the received feedback was very positive, especially in relation to the way the workshops were run and the understanding of equations of state, azeotropes and the various phase diagrams. However, the survey highlighted some shortfalls, such as the students' understanding of activity coefficient models and the retrograde behaviour, which should be looked at in more details. It was also found that the knowledge students have gained in the workshops improved their academic performance in the final exam as the questions concerning phase equilibria, thus closely linked to the workshops, had an average increase of 9.4%, compared to an unrelated question only having an increase of 2.2%.

Looking ahead, the digital workshops might be improved by carrying out binary interaction parameter regressions using experimental VLE data and extending the study to liquid–liquid equilibrium described by isobaric T - x - x diagrams. It should also be considered to create some pre-recorded information before the workshop sessions to demonstrate VLE diagrams in Aspen Properties so that students will have some more general preparation in advance.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Notes on contributors

Mauro Luberti is a Senior Lecturer in Chemical Engineering and UG Programme Director at the University of Manchester following his past role of Consultant Engineer for Siemens. His research focuses on engineering education, adsorption thermodynamics and gas separation processes. He is a Chartered Chemical Engineer, a European Engineer and a Senior Fellow of the Advance Higher Education.

Thomas L. Rodgers is a Reader in Chemical Engineering Education at the University of Manchester and former Head of Education in Chemical Engineering. His research focuses on flexible learning, authentic assessments, and how students prepare for learning activities. He is a Chartered Chemical Engineer, a European Engineer, and a Principal Fellow of the Advance Higher Education.

Percy van der Gryp is an Associate Professor in Chemical Engineering at the University of Pretoria and former Head of Education in Chemical Engineering at the University of Manchester. His research focuses on engineering education, reaction engineering and process design, simulation & modelling. He is a Fellow of the Advance Higher Education.

References

- AspenTech. (2024). www.aspentech.com/en.
- Bale, S., & Singh, R. (2023). Utilizing Simtronics SPM 700 distillation in engineering unit operation laboratory. *Interactive Learning Environments*, 31(5), 3031–3041. doi:10.1080/10494820.2021.1916766
- Bansagi Jr, T., & Rodgers, L. T. (2018). Graphic web-apps for teaching ternary diagrams and liquid–liquid extraction. *Education for Chemical Engineers*, 22, 27–34. doi:10.1016/j.ece.2017.11.002
- Barba, Diego, Brandani, Federico, Capocelli, Mauro, Luberti, Mauro, & Zizza, Arturo. (2015). Process analysis of an industrial waste-to-energy plant: Theory and experiments. *Process Safety and Environmental Protection*, 96, 61–73. doi:10.1016/j.psep.2015.04.007
- Bascunana, J., Leon, S., Gonzalez-Miquel, M., Gonzalez, E. J., & Ramirez, J. (2023). Impact of Jupyter Notebook as a tool to enhance the learning process in chemical engineering modules. *Education for Chemical Engineers*, 44, 155–163. doi:10.1016/j.ece.2023.06.001
- Bell, I. H., Wronski, J., Quoilin, S., & Lemort, V. (2014). Pure and pseudo-pure fluid thermophysical property evaluation and the open-source thermophysical property library CoolProp. *Industrial and Engineering Chemistry Research*, 53(6), 2498–2508. doi:10.1021/ie4033999
- Bergsteiner, H., Avery, G. C., & Neumann, R. (2010). Kolb's experiential learning model: Critique from a modelling perspective. *Studies in Continuing Education*, 32(1), 29–46. doi:10.1080/01580370903534355
- Borreguero, A. M., Valverde, J. L., García-Vargas, J. M., & Sánchez-Silva, L. (2019). Simulator-based learning in the teaching of chemical engineering. *Computer Applications in Engineering Education*, 27(5), 1267–1276. doi:10.1002/cae.22150

- Calvo, L., & Prieto, C. (2016). The teaching of enhanced distillation processes using a commercial simulator and a project-based learning approach. *Education for Chemical Engineers*, 17, 65–74. doi:10.1016/j.ece.2016.07.004
- Carlson, E. C. (1996). Don't gamble with physical properties for simulations. *Chemical Engineering Progress*, 92(4), 35–46.
- Caserta, S., Tomaiuolo, G., & Guido, S. (2021). Use of a smartphone-based student response system in large active-learning chemical engineering thermodynamics classrooms. *Education for Chemical Engineers*, 36, 46–52. doi:10.1016/j.ece.2021.02.003
- Castier, M., & Amer, M. M. (2011). XSEOS: An evolving tool for teaching chemical engineering thermodynamics. *Education for Chemical Engineers*, 6(2), 62–70. doi:10.1016/j.ece.2010.12.002
- Dong, X. (2021). In-class student questions and take-home projects for chemical engineering thermodynamics. *Journal of Chemical Education*, 98(4), 1193–1200. doi:10.1021/acs.jchemed.0c01253
- Elliott, J. R., & Lira, C. T. (2012). *Introductory chemical engineering thermodynamics* (2nd ed.). Prentice Hall.
- Forte, E., Jirasek, F., Bortz, M., Burger, J., Vrabec, J., & Hasse, H. (2019). Digitalization in thermodynamics. *Chemie Ingenieur Technik*, 91(3), 201–214. doi:10.1002/cite.201800056
- Kontogeorgis, G. M., Michelsen, M. L., & Clement, K. H. (2009). Teaching chemical engineering thermodynamics at three levels. *Chemical Engineering Education*, 43(1), 70–78.
- Le, B., Lawrie, G. A., & Wang, J. T. H. (2022). Student self-perception on digital literacy in STEM blended learning environments. *Journal of Science Education and Technology*, 31(3), 303–321. doi:10.1007/s10956-022-09956-1
- Martin, A., Bermejo, M. D., Mato, F. A., & Cocero, M. J. (2011). Teaching advanced equations of state in applied thermodynamics courses using open source programs. *Education for Chemical Engineers*, 6(4), 114–121. doi:10.1016/j.ece.2011.08.003
- Martin, C., Ranalli, J., & Moore, J. (2022). PYromat: A python package for thermodynamic properties. *Journal of Open Source Software*, 7(79), 4757. doi:10.21105/joss.04757
- McCarthy, J. J., & Parker, R. S. (2004). Pillars of chemical engineering: A block-scheduled curriculum. *Chemical Engineering Education*, 38(4), 292–301.
- Norman, G. (2010). Likert scales, levels of measurement and the “laws” of statistics. *Advances in Health Sciences Education*, 15(5), 625–632. doi:10.1007/s10459-010-9222-y
- Paoli, L. T., Inguva, P. K., Haslam, A. J., & Walker, P. J. (2024). Confronting the thermodynamics knowledge gap: A short course on computational thermodynamics in Julia. *Education for Chemical Engineers*, 48, 1–14. doi:10.1016/j.ece.2024.03.002
- Prausnitz, J. M., Lichtenthaler, R. N., & de Azevedo, E. G. (1999). *Molecular thermodynamics of fluid-phase equilibria* (3rd ed.). Prentice Hall.
- Rodgers, L. T., Bolton, P., & van der Gryp, P. (2023). An interactive graph resource for chemical engineering teaching. *Education for Chemical Engineers*, 45, 80–89. doi:10.1016/j.ece.2023.08.004
- Sandler, S. I. (2017). *Chemical, biochemical and engineering thermodynamics* (5th ed.). Wiley.
- Santori, Giulio, & Luberti, Mauro. (2016). Thermodynamics of thermally-driven adsorption compression. *Sustainable Materials and Technologies*, 10, 1–9. doi:10.1016/j.susmat.2016.09.001
- Santori, G., Luberti, M., & Brandani, S. (2015). Common tangent plane in mixed-gas adsorption. *Fluid Phase Equilibria*, 392, 49–55. doi:10.1016/j.fluid.2015.02.017
- Schefflan, R. (2016). *Teach yourself the basics of aspen plus* (2nd ed.). Wiley.
- Smith, J. M., Van Ness, H., Abbott, M., & Swihart, M. (2022). *Introduction to chemical engineering thermodynamics* (9th ed.). McGraw Hill.
- Sullivan, G. M., & Artino, A. R. (2013). Analyzing and interpreting data from Likert-type scales. *Journal of Graduate Medical Education*, 5(4), 541–542. doi:10.4300/JGME-5-4-18
- Suthar, K. J., & Joshipura, M. H. (2023). The integrative approach of learning chemical engineering thermodynamics by using simulation-based exercises. *Education for Chemical Engineers*, 45, 122–129. doi:10.1016/j.ece.2023.09.001
- Wankat, P. C. (2009). The history of chemical engineering and pedagogy. *Chemical Engineering Education*, 43, 216–224.